PROJECT REPORT
CSE0002 [F11]

OPEN SOURCE SOFTWARE -- VLC

Submitted by

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Introduction

Open source software is something that has become very important in today's world of technology. It simply means that the source code (the program code) of the software is available to all, so anyone can see it, use it, and even modify it according to their needs. Because of this, many developers from different parts of the world can contribute and improve the software over time. This helps programmers by making modifications in the original code but not have to make the entire code by themselves, almost creating a community.

One thing I noticed while learning about open source is that it makes things more transparent and flexible. Instead of depending on one company, people can actually understand how a software works and even fix or improve it, according to their programming needs. This is one of the main reasons why open source software is widely used in both industries and education.

In this project, I have chosen VLC Media Player, which is a very commonly used open-source software. VLC is an open source software used by multiple people at some point because it can play nearly all types of audio and video files without any extra setup. It is simple to use but still very powerful in practical usage.

While working on this project, I installed VLC on my Linux system and explored how it behaves in real usage. This helped me understand that open-source software is not just theory but something we interact with regularly.

The aim of this report is to study VLC from different aspects like how it was created, what type of license it uses, and how it works in a Linux system and its overall workings. Along with this, I have also included some shell scripts to understand the practical side of open-source tools and how they can be used in real systems.

PART A- ORGIN AND PHILOSOPHY

Origin

Believe it or not, VLC wasn't some big corporate project. It actually started in the late 90s with a bunch of students at École Centrale Paris. Back then, streaming video over a university network was a total nightmare because existing players were either too buggy or just couldn't handle different formats.

The goal was simple: build something flexible. This eventually turned into the "VideoLAN" project. It's pretty cool because while it started as a campus tool, the students eventually went open-source—which changed everything. Unlike most software we buy today, VLC was born to solve a real-world student problem, not to make a profit.

That's why it's still a powerhouse today. Because developers worldwide can contribute, it's become incredibly stable. Whether you're on Windows, Linux, or a Mac, it just works. That open-source "community" vibe is exactly why it's still the king of media players in 2026.

This shows how open-source projects often start from simple problems but grow into widely used tools. VLC is a perfect example of how community-driven development can create long-lasting software.

What I personally found interesting is that VLC did not start as a commercial product but as a student project solving a real problem. This makes it different from many modern software tools which are often profit-driven from the beginning.

License Analysis

VLC Media Player is released under the GNU General Public License (GPL), which is one of the most commonly used open-source licenses. This license makes sure that this software remains free and open for all users. This is done so that users have complete freedom over the software they use. There are four main freedoms in open-source software. The first freedom is the ability to use the software for any purpose. The second is the freedom to study the software working and modify it according to needs. The third freedom allows users to share the software with others. The fourth freedom is the ability to distribute modified versions of the software.

In the case of VLC, these freedoms are fully supported. Anyone can download VLC, modify its code, and even share their own version of it. However, one important condition of the GPL license is that if someone distributes a modified version, they must also make the source code available under the same license.

When comparing GPL with a more permissive license like MIT, there is a clear difference. The MIT license allows developers to modify the software and even keep their changes private. But what GPL does is that it ensures that all modified versions must remain open-source. I strongly believe that GPL is better for maintaining openness but then MIT gives more flexibility to companies for private data.

Another important concept is “free as in freedom” and not just “free as in free beer.” VLC is free not only in terms of cost but also in terms of user rights. Users are not restricted in how they use or modify the software, which makes it truly open-source.

From my understanding, this makes GLP more suitable for community-driven projects like VLC.

Ethical Reflection

Open source software raises a lot of important questions about how technology should be shared. In my opinion, not all software needs to be open source, but open source plays a very big role in learning and innovation. It allows more people to access knowledge and build new things without starting from zero.

One common question is whether companies should be allowed to earn money using open-source software. Many large companies use open-source tools in their systems.

I think this is fair as long as they follow the license rules and also contribute back to the community in some way.

Another interesting idea is "standing on the shoulders of giants." This means developers can build on work done by others. Open source makes this easier and faster, because people don't need to recreate everything again and again.

At the same time, there should be some balance. Companies also need to protect their own work and investments. So, both open-source and proprietary software have their own importance. In my understanding, open source encourages collaboration, while proprietary software focuses more on control and ownership.

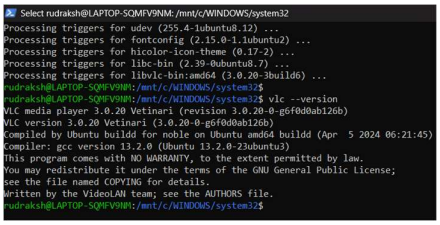
While working on this project, I installed VLC on my Linux system and explored how it behaves in real usage. This helped me understand that open-source software is not just theory but something we interact with regularly.

For example, companies like Google use open source like Linux in Android, but not all parts of systems are open. This creates a debate of whether the companies truly use and support open source or mainly use it for their own benefit.

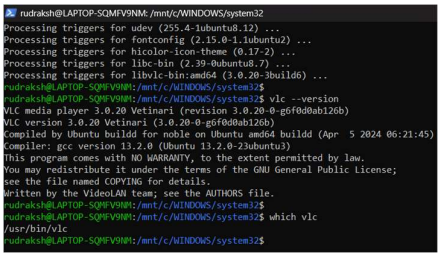
I feel that it is acceptable if they follow the license, but contributing to the community is also equally important.

Part B — Linux Footprint

In this section, VLC Media Player was analysed in a Linux environment to understand how it is installed and managed in the system. VLC can be easily installed using the package manager with the command *sudo apt install vlc*.
After installation, various Linux commands were used to check its details. The command *which vlc* shows the location of the VLC executable file, usually found in */usr/bin/vlc*.
The command *whereis vlc* gives information about related binary and manual files.
Using *dpkg -l | grep vlc*, we can check whether VLC is installed along with its version details. The command *vlc --version* displays the version of VLC currently installed on the system.
The command *ps aux | grep vlc* helps to check running processes related to VLC, while *ls -l /usr/bin/vlc* shows the permissions and ownership of the VLC executable file.
VLC runs as a normal user process, which improves system security. It does not require root privileges to run, which reduces the risk of system-level issues.



VLC version command



which vlc command

```
rudraksh@LAPTOP-SQMFV9NM: /mnt/c/WINDOWS/system32
Processing triggers for udev (255.4-1ubuntu8.12) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for libvlc-bin:amd64 (3.0.20-3build6) ...
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ vlc --version
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
VLC version 3.0.20 Vetinari (3.0.20-0-g6f0d0ab126b)
Compiled by Ubuntu buildd for noble on Ubuntu amd64 buildd (Apr  5 2024 06:21:45)
Compiler: gcc version 13.2.0 (Ubuntu 13.2.0-23ubuntu3)
This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ which vlc
/usr/bin/vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ dpkg -l | grep vlc
ii  libvlc-bin:amd64                3.0.20-3build6                          amd64        tools for VLC's base library
ii  libvlc5:amd64                   3.0.20-3build6                          amd64        multimedia player and streamer library
ii  libvlccore9:amd64               3.0.20-3build6                          amd64        base library for VLC and its modules
ii  vlc                             3.0.20-3build6                          amd64        multimedia player and streamer
ii  vlc-bin                         3.0.20-3build6                          amd64        binaries from VLC
ii  vlc-data                        3.0.20-3build6                          all          common data for VLC
ii  vlc-l10n                        3.0.20-3build6                          all          translations for VLC
ii  vlc-plugin-access-extra:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (extra access plugins)
ii  vlc-plugin-base:amd64           3.0.20-3build6                          amd64        multimedia player and streamer (base plugins)
ii  vlc-plugin-notify:amd64         3.0.20-3build6                          amd64        LibNotify plugin for VLC
ii  vlc-plugin-qt:amd64             3.0.20-3build6                          amd64        multimedia player and streamer (Qt plugin)
ii  vlc-plugin-samba:amd64          3.0.20-3build6                          amd64        Samba plugin for VLC
ii  vlc-plugin-skins2:amd64         3.0.20-3build6                          amd64        multimedia player and streamer (Skins2 plugin)
ii  vlc-plugin-video-output:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (video output plugins)
ii  vlc-plugin-video-splitter:amd64 3.0.20-3build6                          amd64        multimedia player and streamer (video splitter plugins)
ii  vlc-plugin-visualization:amd64  3.0.20-3build6                          amd64        multimedia player and streamer (visualization plugins)
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
```

vlc packages commands

```
rudraksh@LAPTOP-SQMFV9NM: /mnt/c/WINDOWS/system32
Processing triggers for udev (255.4-1ubuntu8.12) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for libvlc-bin:amd64 (3.0.20-3build6) ...
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ vlc --version
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
VLC version 3.0.20 Vetinari (3.0.20-0-g6f0d0ab126b)
Compiled by Ubuntu buildd for noble on Ubuntu amd64 buildd (Apr  5 2024 06:21:45)
Compiler: gcc version 13.2.0 (Ubuntu 13.2.0-23ubuntu3)
This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ which vlc
/usr/bin/vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ dpkg -l | grep vlc
ii  libvlc-bin:amd64                3.0.20-3build6                          amd64        tools for VLC's base library
ii  libvlc5:amd64                   3.0.20-3build6                          amd64        multimedia player and streamer library
ii  libvlccore9:amd64               3.0.20-3build6                          amd64        base library for VLC and its modules
ii  vlc                             3.0.20-3build6                          amd64        multimedia player and streamer
ii  vlc-bin                         3.0.20-3build6                          amd64        binaries from VLC
ii  vlc-data                        3.0.20-3build6                          all          common data for VLC
ii  vlc-l10n                        3.0.20-3build6                          all          translations for VLC
ii  vlc-plugin-access-extra:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (extra access plugins)
ii  vlc-plugin-base:amd64           3.0.20-3build6                          amd64        multimedia player and streamer (base plugins)
ii  vlc-plugin-notify:amd64         3.0.20-3build6                          amd64        LibNotify plugin for VLC
ii  vlc-plugin-qt:amd64             3.0.20-3build6                          amd64        multimedia player and streamer (Qt plugin)
ii  vlc-plugin-samba:amd64          3.0.20-3build6                          amd64        Samba plugin for VLC
ii  vlc-plugin-skins2:amd64         3.0.20-3build6                          amd64        multimedia player and streamer (Skins2 plugin)
ii  vlc-plugin-video-output:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (video output plugins)
ii  vlc-plugin-video-splitter:amd64 3.0.20-3build6                          amd64        multimedia player and streamer (video splitter plugins)
ii  vlc-plugin-visualization:amd64  3.0.20-3build6                          amd64        multimedia player and streamer (visualization plugins)
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ whereis vlc
vlc: /usr/bin/vlc /usr/lib/x86_64-linux-gnu/vlc /usr/share/vlc /usr/share/man/man1/vlc.1.gz
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
```

whereis vlc command

```
rudraksh@LAPTOP-SQMFV9NM: /mnt/c/WINDOWS/system32
Processing triggers for udev (255.4-1ubuntu8.12) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for libvlc-bin:amd64 (3.0.20-3build6) ...
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rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ vlc --version
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You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
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rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ which vlc
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ii  vlc-bin                         3.0.20-3build6                          amd64        binaries from VLC
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ii  vlc-l10n                        3.0.20-3build6                          all          translations for VLC
ii  vlc-plugin-access-extra:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (extra access plugins)
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ii  vlc-plugin-notify:amd64         3.0.20-3build6                          amd64        LibNotify plugin for VLC
ii  vlc-plugin-qt:amd64             3.0.20-3build6                          amd64        multimedia player and streamer (Qt plugin)
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ii  vlc-plugin-video-splitter:amd64 3.0.20-3build6                          amd64        multimedia player and streamer (video splitter plugins)
ii  vlc-plugin-visualization:amd64  3.0.20-3build6                          amd64        multimedia player and streamer (visualization plugins)
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ whereis vlc
vlc: /usr/bin/vlc /usr/lib/x86_64-linux-gnu/vlc /usr/share/vlc /usr/share/man/man1/vlc.1.gz
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ ps aux | grep vlc
rudraksh   3909  0.0  0.0   4088  1920 pts/0    S+   18:01   0:00 grep --color=auto vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
```

vlc process commands

```
rudraksh@LAPTOP-SQMFV9NM: /mnt/c/WINDOWS/system32
Processing triggers for udev (255.4-1ubuntu8.12) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for libvlc-bin:amd64 (3.0.20-3build6) ...
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ vlc --version
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
VLC version 3.0.20 Vetinari (3.0.20-0-g6f0d0ab126b)
Compiled by Ubuntu buildd for noble on Ubuntu amd64 buildd (Apr  5 2024 06:21:45)
Compiler: gcc version 13.2.0 (Ubuntu 13.2.0-23ubuntu3)
This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
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rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ which vlc
/usr/bin/vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ dpkg -l | grep vlc
ii  libvlc-bin:amd64                3.0.20-3build6                          amd64        tools for VLC's base library
ii  libvlc5:amd64                   3.0.20-3build6                          amd64        multimedia player and streamer library
ii  libvlccore9:amd64               3.0.20-3build6                          amd64        base library for VLC and its modules
ii  vlc                             3.0.20-3build6                          amd64        multimedia player and streamer
ii  vlc-bin                         3.0.20-3build6                          amd64        binaries from VLC
ii  vlc-data                        3.0.20-3build6                          all          common data for VLC
ii  vlc-l10n                        3.0.20-3build6                          all          translations for VLC
ii  vlc-plugin-access-extra:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (extra access plugins)
ii  vlc-plugin-base:amd64           3.0.20-3build6                          amd64        multimedia player and streamer (base plugins)
ii  vlc-plugin-notify:amd64         3.0.20-3build6                          amd64        LibNotify plugin for VLC
ii  vlc-plugin-qt:amd64             3.0.20-3build6                          amd64        multimedia player and streamer (Qt plugin)
ii  vlc-plugin-samba:amd64          3.0.20-3build6                          amd64        Samba plugin for VLC
ii  vlc-plugin-skins2:amd64         3.0.20-3build6                          amd64        multimedia player and streamer (Skins2 plugin)
ii  vlc-plugin-video-output:amd64   3.0.20-3build6                          amd64        multimedia player and streamer (video output plugins)
ii  vlc-plugin-video-splitter:amd64 3.0.20-3build6                          amd64        multimedia player and streamer (video splitter plugins)
ii  vlc-plugin-visualization:amd64  3.0.20-3build6                          amd64        multimedia player and streamer (visualization plugins)
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ whereis vlc
vlc: /usr/bin/vlc /usr/lib/x86_64-linux-gnu/vlc /usr/share/vlc /usr/share/man/man1/vlc.1.gz
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ ps aux | grep vlc
rudraksh    3909  0.0  0.0   4088  1920 pts/0    S+   18:01   0:00 grep --color=auto vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$ ls -l /usr/bin/vlc
-rwxr-xr-x 1 root root 14640 Apr  5  2024 /usr/bin/vlc
rudraksh@LAPTOP-SQMFV9NM:/mnt/c/WINDOWS/system32$
```

vlc permissions commands

User and Group:
VLC runs as a normal user rather than a root user. This improves system security because it limits access to critical system files and prevents accidental system damage.

Service Management:
VLC is not a background service. It runs only when the user launches it, either through the terminal or graphical interface. This makes it lightweight and efficient.

Update Model:
VLC is updated using the Linux package manager. Commands like sudo apt update and sudo apt upgrade ensure that the latest updates and security patches are installed.

Part C- FOSS Ecosystem

VLC Media Player is not an isolated software; it is part of a larger open-source ecosystem. It depends on several open-source libraries and tools to function effectively. One of the most important dependencies is FFmpeg, which is used for decoding and encoding different audio and video formats.

In addition to this, VLC uses various codecs and multimedia libraries that allow it to support a wide range of media files. Because of these dependencies, VLC is able to provide a seamless experience without requiring users to install additional components manually.

VLC has also enabled many features such as media streaming, format conversion, and cross-platform playback. It is widely used in different real world scenarios and environments including personal systems, educational institutions, and even broadcasting.

The project is maintained by the VideoLAN community, which includes developers from around the world. This community actively contributes to bug fixes, new features, and updates, ensuring that the software remains reliable and up to date.

Because of this ecosystem, VLC continues to evolve and adapt to new media technologies. This ensures its relevance even as media formats change over time.

PART D- Comparison with Proprietary Software

Feature	VLC	Proprietary Player
Cost	Free	Paid
Source Code	Open	Closed
Flexibility	High	Limited
Support	Community	Company
Customisation	Full	Limited
Updates	Frequent	Controlled
Security	Transparent	Opaque
Control	User	Company
Platform	Cross-platform	Restricted
License	GPL	Proprietary

From this comparison, it is clear that VLC provides more flexibility and control to users because it is open-source. Users are free to modify, distribute, and customize the software according to their needs. This makes it highly suitable for a wide range of applications.

On the other hand, proprietary software may offer better official support and a more polished user experience. However, users have limited control over how the software works and cannot modify it.

In my opinion, VLC is a better choice for most general users because it is free and flexible. It clearly demonstrates how open-source software can compete with and even outperform proprietary alternatives in real-world usage.

SHELL SCRIPTS TASKS

1) Script 1- System Identity Report

Script 1 Program Code:

```
rudraksh@LAPTOP-SQMFV9N
rudraksh@LAPTOP-SQMFV9NM:~$ cat script1.sh
#!/bin/bash
# Script 1: System Identity Report
# Author: Rudraksh Mehta
# Description: Displays system information

echo "===================================="
echo " Open Source Audit System Report"
echo "===================================="

# Display student name
echo "Student Name: Rudraksh Mehta"

# Get system details
USER_NAME=$(whoami)
DATE=$(date)
KERNEL=$(uname -r)
UPTIME=$(uptime -p)
HOME_DIR=$HOME

# Display details
echo "User: $USER_NAME"
echo "Home Directory: $HOME_DIR"
echo "Date & Time: $DATE"
echo "Kernel Version: $KERNEL"
echo "System Uptime: $UPTIME"

# License message
echo "License: GNU General Public License (GPL)"

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$
```

Explanation:

This script displays basic system information such as user name, kernel version, uptime, and date. It uses variables and command substitution to collect and print the data.

Concepts used:

variables, command substitution, echo formatting

Script 1 Output:

```
rudraksh@LAPTOP-SQMFV9N

rudraksh@LAPTOP-SQMFV9NM:~$ cat script1.sh
#!/bin/bash
# Script 1: System Identity Report
# Author: Rudraksh Mehta
# Description: Displays system information

echo "====================================="
echo " Open Source Audit System Report"
echo "====================================="

# Display student name
echo "Student Name: Rudraksh Mehta"

# Get system details
USER_NAME=$(whoami)
DATE=$(date)
KERNEL=$(uname -r)
UPTIME=$(uptime -p)
HOME_DIR=$HOME

# Display details
echo "User: $USER_NAME"
echo "Home Directory: $HOME_DIR"
echo "Date & Time: $DATE"
echo "Kernel Version: $KERNEL"
echo "System Uptime: $UPTIME"

# License message
echo "License: GNU General Public License (GPL)"

echo "====================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script1.sh
=====================================
 Open Source Audit System Report
=====================================
Student Name: Rudraksh Mehta
User: rudraksh
Home Directory: /home/rudraksh
Date & Time: Tue Mar 31 14:00:48 IST 2026
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
System Uptime: up 9 minutes
License: GNU General Public License (GPL)
=====================================
rudraksh@LAPTOP-SQMFV9NM:~$
```

2) Script 2-FOSS Package Inspector

Script 2 Program Code:

```
rudraksh@LAPTOP-SQMFV9N
rudraksh@LAPTOP-SQMFV9NM:~$ cat script2.sh
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Rudraksh Mehta
# Description: Checks if VLC is installed in Linux and shows details

PACKAGE="vlc"

echo "===================================="
echo "   FOSS Package Inspection Report   "
echo "===================================="

echo "Checking package: $PACKAGE"
echo ""

# Check installation using command existence
if command -v $PACKAGE &> /dev/null; then
    echo "$PACKAGE is installed in this Linux environment."

    # Show version (first 2 lines only)
    $PACKAGE --version | head -n 2
else
    echo "$PACKAGE is NOT installed in this Linux environment."
    echo "Note: It may be installed on Windows, but not inside WSL Ubuntu."
    echo "Suggestion: Install using 'sudo apt install vlc'"
fi

echo ""

# Case statement for description
case $PACKAGE in
    vlc)
        echo "VLC: Open-source multimedia player that supports almost all media formats."
        ;;
    firefox)
        echo "Firefox: Open-source web browser focused on privacy."
        ;;
    *)
        echo "Unknown package."
        ;;
esac

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$
```

Script 2 Output:

```
rudraksh@LAPTOP-SQMFV9N

echo "Checking package: $PACKAGE"
echo ""

# Check installation using command existence
if command -v $PACKAGE &> /dev/null; then
    echo "$PACKAGE is installed in this Linux environment."

    # Show version (first 2 lines only)
    $PACKAGE --version | head -n 2
else
    echo "$PACKAGE is NOT installed in this Linux environment."
    echo "Note: It may be installed on Windows, but not inside WSL Ubuntu."
    echo "Suggestion: Install using 'sudo apt install vlc'"
fi

echo ""

# Case statement for description
case $PACKAGE in
    vlc)
        echo "VLC: Open-source multimedia player that supports almost all media formats."
        ;;
    firefox)
        echo "Firefox: Open-source web browser focused on privacy."
        ;;
    *)
        echo "Unknown package."
        ;;
esac

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script2.sh
====================================
   FOSS Package Inspection Report
====================================
Checking package: vlc

vlc is NOT installed in this Linux environment.
Note: It may be installed on Windows, but not inside WSL Ubuntu.
Suggestion: Install using 'sudo apt install vlc'

VLC: Open-source multimedia player that supports almost all media formats.
====================================
rudraksh@LAPTOP-SQMFV9NM:~$
```

Explanation:

This script checks whether a package is installed in the system and displays its details. It uses if-else conditions, grep, and a case statement.

Concepts used: if-else, grep, case statement

3) Script 3-Disk and Permission Auditor

Script 3 Program Code:

```
rudraksh@LAPTOP-SQMFV9N:~$ cat script3.sh
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: Rudraksh Mehta
# Description: Displays permissions, owner, and size of important directories

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "====================================="
echo "   Directory Audit Report            "
echo "====================================="

for DIR in "${DIRS[@]}"
do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1}')
        OWNER=$(ls -ld $DIR | awk '{print $3}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

        echo "Directory: $DIR"
        echo "Permissions: $PERMS"
        echo "Owner: $OWNER"
        echo "Size: $SIZE"
        echo "-----------------------------------"
    else
        echo "$DIR does not exist"
    fi
done

echo "====================================="
rudraksh@LAPTOP-SQMFV9NM:~$
```

Explanation:

This script checks important directories in the system and shows their permissions and size. It uses loops, conditions, and commands like ls and du.

Concepts used: loops, conditions, awk, file handling

Script 3 Output:

```
rudraksh@LAPTOP-SQMFV9N

        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

        echo "Directory: $DIR"
        echo "Permissions: $PERMS"
        echo "Owner: $OWNER"
        echo "Size: $SIZE"
        echo "-----------------------------------"
    else
        echo "$DIR does not exist"
    fi
done

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script3.sh
====================================
   Directory Audit Report
====================================
Directory: /etc
Permissions: drwxr-xr-x
Owner: root
Size: 4.1M
-----------------------------------
Directory: /var/log
Permissions: drwxrwxr-x
Owner: root
Size: 802M
-----------------------------------
Directory: /home
Permissions: drwxr-xr-x
Owner: root
Size: 404K
-----------------------------------
Directory: /usr/bin
Permissions: drwxr-xr-x
Owner: root
Size: 122M
-----------------------------------
Directory: /tmp
Permissions: drwxrwxrwt
Owner: root
Size: 20K
-----------------------------------
====================================
rudraksh@LAPTOP-SQMFV9NM:~$
```

4) Script 4- Log File Analyser

Script 4 Program Code:

```
rudraksh@LAPTOP-SQMFV9N
rudraksh@LAPTOP-SQMFV9NM:~$ cat script4.sh
#!/bin/bash
# Script 4: Log File Analyzer
# Author: Rudraksh Mehta

LOGFILE="/var/log/dpkg.log"
KEYWORD="install"

echo "====================================="
echo "   Log File Analysis Report         "
echo "====================================="

echo "Analyzing file: $LOGFILE"
echo ""

if [ ! -f "$LOGFILE" ]; then
    echo "Log file not found!"
    exit 1
fi

# Count occurrences (fast method)
COUNT=$(grep -i "$KEYWORD" "$LOGFILE" | wc -l)

echo "Keyword '$KEYWORD' found $COUNT times."

echo ""
echo "Last 5 matching entries:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5

echo "====================================="
rudraksh@LAPTOP-SQMFV9NM:~$
```

Script 4 Output:

```
rudraksh@LAPTOP-SQMFV9N

# Script 4: Log File Analyzer
# Author: Rudraksh Mehta

LOGFILE="/var/log/dpkg.log"
KEYWORD="install"

echo "===================================="
echo "   Log File Analysis Report         "
echo "===================================="

echo "Analyzing file: $LOGFILE"
echo ""

if [ ! -f "$LOGFILE" ]; then
    echo "Log file not found!"
    exit 1
fi

# Count occurrences (fast method)
COUNT=$(grep -i "$KEYWORD" "$LOGFILE" | wc -l)

echo "Keyword '$KEYWORD' found $COUNT times."

echo ""
echo "Last 5 matching entries:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script4.sh
====================================
   Log File Analysis Report
====================================
Analyzing file: /var/log/dpkg.log

Keyword 'install' found 2091 times.

Last 5 matching entries:
2025-09-09 07:15:15 status installed g++-11:amd64 11.4.0-1ubuntu1~22.04.2
2025-09-09 07:15:15 status installed g++:amd64 4:11.2.0-1ubuntu1
2025-09-09 07:15:15 status installed build-essential:amd64 12.9ubuntu3
2025-09-09 07:15:16 status installed man-db:amd64 2.10.2-1
2025-09-09 07:15:16 status installed libc-bin:amd64 2.35-0ubuntu3.8
====================================
rudraksh@LAPTOP-SQMFV9NM:~$
```

Explanation:

This script reads a log file and counts how many times a specific keyword appears. It uses a while loop, conditions, and basic text processing.

Concepts used: while loop, conditions, string matching

5) Script 5- Open Source Manifesto Generator

Script 5 Program Code:

```
rudraksh@LAPTOP-SQMFV9N
rudraksh@LAPTOP-SQMFV9NM:~$ cat script5.sh
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Rudraksh Mehta
# Description: Generates a personalized open source manifesto

echo "===================================="
echo "   Open Source Manifesto Generator  "
echo "===================================="

echo ""

read -p "1. Name one open-source tool you use daily: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date)
OUTPUT="manifesto.txt"

echo ""
echo "Generating your manifesto..."
echo ""

# Writing to file
echo "On $DATE, I believe in the power of open source." > $OUTPUT
echo "I use $TOOL regularly in my daily work." >> $OUTPUT
echo "To me, freedom means $FREEDOM." >> $OUTPUT
echo "I aspire to build $BUILD and share it with the world." >> $OUTPUT
echo "Open source enables collaboration, innovation, and transparency." >> $OUTPUT

echo "Manifesto saved in file: $OUTPUT"
echo ""

echo "----- Your Manifesto -----"
cat $OUTPUT

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$
```

Script 5 Output:

```
rudraksh@LAPTOP-SQMFV9N

read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date)
OUTPUT="manifesto.txt"

echo ""
echo "Generating your manifesto..."
echo ""

# Writing to file
echo "On $DATE, I believe in the power of open source." > $OUTPUT
echo "I use $TOOL regularly in my daily work." >> $OUTPUT
echo "To me, freedom means $FREEDOM." >> $OUTPUT
echo "I aspire to build $BUILD and share it with the world." >> $OUTPUT
echo "Open source enables collaboration, innovation, and transparency." >> $OUTPUT

echo "Manifesto saved in file: $OUTPUT"
echo ""

echo "----- Your Manifesto -----"
cat $OUTPUT

echo "==================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script5.sh
===================================
   Open Source Manifesto Generator
===================================

1. Name one open-source tool you use daily: VLC MEDIA PLAYER
2. In one word, what does 'freedom' mean to you? INNOVATION
3. Name one thing you would build and share freely: OPEN SOURCE TOOLS

Generating your manifesto...

Manifesto saved in file: manifesto.txt

----- Your Manifesto -----
On Tue Mar 31 14:32:22 IST 2026, I believe in the power of open source.
I use VLC MEDIA PLAYER regularly in my daily work.
To me, freedom means INNOVATION.
I aspire to build OPEN SOURCE TOOLS and share it with the world.
Open source enables collaboration, innovation, and transparency.
===================================
rudraksh@LAPTOP-SQMFV9NM:~$
```

```
rudraksh@LAPTOP-SQMFV9N

echo ""

# Writing to file
echo "On $DATE, I believe in the power of open source." > $OUTPUT
echo "I use $TOOL regularly in my daily work." >> $OUTPUT
echo "To me, freedom means $FREEDOM." >> $OUTPUT
echo "I aspire to build $BUILD and share it with the world." >> $OUTPUT
echo "Open source enables collaboration, innovation, and transparency." >> $OUTPUT

echo "Manifesto saved in file: $OUTPUT"
echo ""

echo "----- Your Manifesto -----"
cat $OUTPUT

echo "===================================="
rudraksh@LAPTOP-SQMFV9NM:~$ ./script5.sh
====================================
   Open Source Manifesto Generator
====================================

1. Name one open-source tool you use daily: VLC MEDIA PLAYER
2. In one word, what does 'freedom' mean to you? INNOVATION
3. Name one thing you would build and share freely: OPEN SOURCE TOOLS

Generating your manifesto...

Manifesto saved in file: manifesto.txt

----- Your Manifesto -----
On Tue Mar 31 14:32:22 IST 2026, I believe in the power of open source.
I use VLC MEDIA PLAYER regularly in my daily work.
To me, freedom means INNOVATION.
I aspire to build OPEN SOURCE TOOLS and share it with the world.
Open source enables collaboration, innovation, and transparency.
====================================
rudraksh@LAPTOP-SQMFV9NM:~$ cat manifesto.txt
On Tue Mar 31 14:32:22 IST 2026, I believe in the power of open source.
I use VLC MEDIA PLAYER regularly in my daily work.
To me, freedom means INNOVATION.
I aspire to build OPEN SOURCE TOOLS and share it with the world.
Open source enables collaboration, innovation, and transparency.
rudraksh@LAPTOP-SQMFV9NM:~$
```

Explanation:
This script takes input from the user and generates a small text file based on the answers. It uses read statements, variables, and file handling.

Concepts used: user input, variables, file writing

REFERENCES

1. https://www.videolan.org/
2. https://www.gnu.org/licenses/gpl-3.0.en.html
3. Ubuntu Documentation
4. Linux Manual Pages